

Andisa khumalo

South Africa, Richards Bay | andisakhumalo.94@gmail.com | 0740139366 | www.linkedin.com/in/andisa-khumalo-186761235

PERSONAL PROFILE

Final-year Electrical and Computer Engineering student at University of Zululand, with hands-on experience spanning industrial automation, embedded systems, and full-stack web development. Seeking a position in the summer university hosted by Ural Federal University where I can apply a strong technical foundation built across ASP.NET Core, and AI-integrated hardware to solve real-world engineering challenges.

EDUCATION

University of Engineering | *B.ENG(Electrical and Computer)*

Richards Bay | **2023– Present**

- **Grade:** First Class Honours
- **Key Modules:**
 - Software Engineering: Object-oriented design, full-stack web development, database management, and software development lifecycles using C#, ASP.NET Core MVC, and SQL Server.
 - Assembly Language & Operating Systems: Low-level programming, memory management, process scheduling, and hardware-software interfacing on embedded and desktop platforms.
 - Process Control & Instrumentation: PID control design, industrial instrumentation, SCADA systems, and IEC 61131-3 PLC programming applied to real-world automation scenarios.
 - Embedded Systems: Microcontroller programming, sensor integration, real-time control, and hardware-software co-design for autonomous and IoT applications.
 - Signals & Systems: Signal processing theory, Fourier and Laplace transforms, filter design, and system modelling applied to communications and control.
 - Communications & Networks: Data transmission, network protocols, IoT communication architectures, and RFID-based real-time data pipeline design.

Secondary School | *Qhamuka High School*

Empangeni | **2017 - 2021**

- 7 subjects including Maths, Physical Sciences and English

UNIVERSITY/PERSONAL PROJECTS

Project 1

AK Mechanics — Vehicle Service Tracker

2026

- Designed and built a full-stack web application for vehicle service and maintenance management, serving as sole developer from requirements through to live deployment.
- Demonstrated proficiency in ASP.NET Core MVC, Entity Framework Core, SQL Server, and ASP.NET Identity for role-based authentication and authorisation.
- Deployed the live application to Microsoft Azure, achieving full end-to-end functionality including user management, service records, and reporting dashboards

AI Autonomous RC Car

2026- Present

- Developed an autonomous ground vehicle integrating computer vision and embedded machine learning for real-time obstacle avoidance and lane following.
- Demonstrated skills in Python, sensor fusion, and embedded hardware integration, managing the full project lifecycle from design through testing independently.
- Successfully achieved reliable autonomous navigation at low speed, validating the control algorithm through structured physical testing.

ACTIVITIES & MEMBERSHIPS

Competitor — Huawei ICT Competition

2025 – Present

South Africa Regional Round

- Currently competing in the Huawei ICT Competition, applying knowledge in networking, cloud computing, and emerging technologies in a structured competitive environment.
- Demonstrates initiative, self-directed learning, and commitment to professional development beyond academic requirements.
- Develops skills in technical problem-solving, time management, and performance under assessment conditions.

Student Member — SAIEE

2024 – Present

South African Institute of Electrical Engineers

- Active student member of the SAIEE, engaging with the broader engineering professional community and staying current with industry developments.
- Demonstrates professional awareness and commitment to the engineering profession from an early career stage.

SKILLS

Programming Languages:

C#, Python, C/C++, MATLAB = Proficient

Web Development:

ASP.NET Core MVC, HTML/CSS, Bootstrap 5, SQL Server = Proficient

Industrial Automation:

PLC Programming (IEC 61131-3 Structured Text), SCADA, HVAC Control = Proficient

Embedded Systems & Hardware:

Microcontrollers, Sensor Fusion, PCB/Circuit Design, RFID = Intermediate

Cloud & DevOps:

Microsoft Azure, Entity Framework Core, Git/GitHub = Intermediate

Simulation & Modelling:

MATLAB/Simulink, Computer Vision, PID Design = Intermediate

INTERESTS

- Football: Regular participant in competitive 5-a-side and 11-a-side football, demonstrating teamwork, communication, and performance under pressure.
- Engineering Simulation: Enjoys building MATLAB/Simulink models and control system simulations in personal time, reflecting a genuine passion for applied engineering.

REFERENCES

References are available upon request